Block One Synthesis: The Pediatric Gastroenterologist's Toolkit.

Imagine a single clinic morning. A three-year-old boy has had loose stools for two months, and his fecal calprotectin is three hundred and eighty micrograms per gram. A nine-year-old girl arrives with two breath test reports, one labeled fructose malabsorption and the other labeled small intestinal bacterial overgrowth. A seven-month-old with Down syndrome has been constipated since the second week of life, his contrast enema was called normal, and a manometry study did not show the reflex that should be there. A fourteen-year-old girl with falling height has a coeliac antibody twelve times the upper limit of normal, and her family is moving to another country. And a fourteen-month-old boy has bloody diarrhea, a fistula next to his anus, and recurrent skin abscesses.

Every one of these children has already had a test. The clinical question of this chapter, which closes the first block of the course, is not which test exists, but which test answers which question, in what order, and what could fool it. The main idea of the whole block fits in one sentence. Every tool a pediatric gastroenterologist uses answers one question well and the others badly, so the skill is choosing the order, not knowing the menu. Before every order, ask three things: what exactly is the question, what would change if the answer came back positive or negative, and what could make this particular test lie in this particular child.

A note on sources. This chapter draws together the thirteen earlier lessons of the block and three anchor documents. The first is the 2017 guideline on pediatric gastrointestinal endoscopy written jointly by the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN, and the European Society of Gastrointestinal Endoscopy, known as ESGE, published by Thomson, Tringali and colleagues. The second is the set of quality standards published in 2022 by the Pediatric Endoscopy Quality Improvement Network, known as PEnQuIN, a joint project of ESPGHAN and the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, led by Walsh and Lightdale. The third is the 2017 North American consensus on hydrogen and methane breath testing, led by Rezaie. Along the way we also use the 2012 consensus on combined pH and impedance monitoring in children by Wenzl and colleagues, and the 2014 pediatric malnutrition consensus of the American Society for Parenteral and Enteral Nutrition and the Academy of Nutrition and Dietetics, led by Becker.

Part one. Four families of tools.

It helps to sort the block into four families. The first family is growth and history. Its lessons were that a trajectory beats a single point, that growth should be read as z-scores, meaning the

number of standard deviations a child sits above or below the reference mean, rather than as centiles, and that alarm features direct which tests to send rather than establishing a diagnosis. The second family is blood and stool. Calprotectin triages; normal blood tests exclude nothing; and a positive result on a multiplex stool test, which looks for the genetic material of many organisms at once, is a detection rather than proof of a cause. The third family is physiology tests: breath tests, combined pH and impedance monitoring, and manometry, which measures pressures in the gut. Here the lesson was that adult normal values mislabel children, and that fermentation is not a disease. The fourth family is the scope, the tissue, the image and the gene. Biopsy the normal-looking mucosa, prefer ultrasound and magnetic resonance enterography to computed tomography, and sequence early when the onset of disease is early.

Part two. Every test has a blind spot.

The most useful way to remember these tools is by what fools them. Fecal calprotectin measures a protein released by neutrophils, the white cells of acute inflammation, into the gut lumen. It is fooled by young age, because healthy children under about four years have higher values, and by non-steroidal anti-inflammatory drugs, proton pump inhibitors and infection. It can never tell you which disease is present. Fecal elastase measures pancreatic enzyme output, and it is fooled by watery stool, which dilutes it, and by small-bowel disease such as coeliac disease, which lowers it without pancreatic failure. A multiplex stool test detects nucleic acid, and it is fooled by carriage and by shedding after an illness has resolved; it cannot tell you whether the organism caused the symptoms.

The breath test measures fermentation, either by colonic bacteria or by bacteria that have moved up into the small bowel. It is fooled by rapid transit, which delivers the sugar to the colon early, and by children whose bacteria make methane instead of hydrogen. It cannot reliably diagnose small-bowel overgrowth. Combined pH and impedance monitoring counts reflux events and measures acid exposure, but there are no true pediatric normal values, and the test cannot tell you on its own whether reflux causes the symptoms. High-resolution manometry measures pressure and propagation, and adult thresholds mislabel small children. Anorectal manometry tests whether the internal anal sphincter relaxes when the rectum is distended, a reflex called the rectoanal inhibitory reflex, or RAIR. Anesthesia and a small rectum can distort it, and it can never show you ganglion cells. Endoscopy with biopsy shows mucosal architecture and cells, but only if you biopsy beyond the visible lesions, and it cannot see disease deeper than the mucosa. Intestinal ultrasound and magnetic resonance enterography, called MRE, measure wall thickness and blood flow, and they are fooled by adult cut-offs. And a gene panel finds variants only in the genes chosen, it produces variants of uncertain significance, often called VUS, and it is silent about every gene not on the list.

Part three. The anchor documents.

The 2017 ESPGHAN and ESGE endoscopy guideline covers diagnostic and therapeutic endoscopy from birth to eighteen years, including foreign bodies, caustic ingestion, bleeding, and endoscopic retrograde cholangiopancreatography and endoscopic ultrasound, but not gastrostomy or inflammatory bowel disease endoscopy, which have their own documents. It used the GRADE system, and most of its statements are weak recommendations based on low-quality evidence. Two lines are worth quoting from memory. First, upper endoscopy and ileocolonoscopy in children are suggested under general anesthesia, or under deep sedation only if general anesthesia is not available. Second, routine tissue sampling is suggested even when the mucosa looks normal. The same guideline also states that upper endoscopy is not suggested for uncomplicated reflux or functional disorders, that low-volume bowel preparation is recommended and sodium phosphate preparations are recommended against, and that button batteries impacted in the esophagus must be removed within two hours.

PEnQuIN, published in 2022, reached consensus among members from thirty-one centers in eleven countries on forty-nine quality standards and forty-seven indicators, covering facilities, procedures, endoscopists and the patient experience. The evidence was mostly adult and observational and rated very low in quality, so forty-five of the forty-nine standards are conditional. The numbers to carry are these: adequate bowel preparation in at least eighty percent of colonoscopies, reaching the cecum in at least ninety percent, and intubating the terminal ileum in at least eighty-five percent.

The North American breath testing consensus brought together expert clinicians who drafted twenty-eight statements, twenty-six of which reached agreement. Its key numbers are a hydrogen rise of at least twenty parts per million above baseline within ninety minutes as a positive test for small-bowel overgrowth, methane of at least ten parts per million as positive, standard doses of ten grams of lactulose, seventy-five grams of glucose, and twenty-five grams each of lactose and fructose, avoidance of antibiotics for four weeks before testing, and no need to stop proton pump inhibitors. It also said that breath tests should not be used to measure transit from mouth to cecum. Importantly for us, it gave no pediatric doses and listed them as a knowledge gap.

The 2012 pH and impedance consensus described how to perform and read the study in children, and both it and the later reflux guideline accept that true pediatric normal values do not exist, because healthy children cannot ethically be catheterized for research. And the 2014 malnutrition consensus established that a single indicator is enough to diagnose malnutrition, graded as mild, moderate or severe by z-scores of minus one, minus two and minus three.

Part four. Where the guidelines agree and disagree.

In every lesson of this block we compared North American and European documents and gave each question one of four verdicts: aligned, differing in detail, materially different, or a gap where no document answers the question. Across the thirteen lessons there were ninety-three such questions. Forty-three, or forty-six percent, were aligned. Twenty-four, about a quarter, differed in detail. Thirteen, or fourteen percent, were materially different, and another thirteen were gaps. So about half the time the two sides of the Atlantic give the same answer. That is noticeably better than the hepatology toolkit block, where only about twenty-eight percent of questions were aligned, probably because PEnQuIN, the reflux guideline and several other documents were written jointly by the two pediatric societies. Almost every material difference came from one of two sources: applying adult numbers to children, or differing policies on radiation and anesthesia.

Look first at the scope and the tissue. Both sides use anesthesia-supported endoscopy in most centers, and the European guideline prefers general anesthesia. Clear fluids are allowed until two hours before anesthesia under the American Society of Anesthesiologists, and until one hour before under the European anesthesia society, a difference in detail. The biopsy protocols are aligned: at least six esophageal biopsies from at least two levels for eosinophilic esophagitis, and four distal duodenal biopsies plus the bulb for coeliac disease. A patency capsule before video capsule endoscopy in known Crohn disease is a strong recommendation in the 2023 European update but has no pediatric mandate in North America. Intestinal ultrasound is first-line in the 2025 European imaging guidance but only an adjunct in North America. And neither side has a quality indicator for whether the right number of biopsies was taken.

Two differences are material. The first is coeliac disease without biopsy. Under the 2020 ESPGHAN guideline, a child with a tissue transglutaminase IgA antibody at least ten times the upper limit of normal and a positive endomysial antibody on a second blood sample can be diagnosed without endoscopy. The 2016 NASPGHAN guidance still advised biopsy, and the 2023 American College of Gastroenterology guideline accepts the no-biopsy route only in children and only conditionally. So the same child gets an endoscopy in one system and not in the other. The second is computed tomography enterography, which the American College of Radiology still rates as usually appropriate in children, whereas European practice reserves it for emergencies because of radiation.

Now the physiology tests. Both sides agree that breath tests should not be used to measure transit. They differ on whether a hydrogen rise defines malabsorption or whether symptoms are also needed before calling it intolerance. They differ on acid exposure: the joint 2018 pediatric reflux guideline quotes thresholds of about ten percent in infants and five percent in older children, while the adult Lyon 2.0 consensus calls under four percent normal and over six

percent abnormal. Four questions are material. There are no agreed pediatric breath test doses. The adult Chicago classification of esophageal motility over-calls disease in children. Adult London criteria for dyssynergic defecation, the failure to relax the pelvic floor while pushing, label many healthy children as abnormal. And for Clostridioides difficile, the 2017 pediatric recommendation of metronidazole or vancomycin has been overtaken in adults by fidaxomicin, which the 2021 updates prefer.

Part five. Sensitivity, test by test.

It is worth putting the main diagnostic numbers of the block side by side, because they change how hard you lean on a result. Ultrasound for intussusception had a sensitivity of about ninety-eight percent, with a negative predictive value of almost one hundred percent. Ultrasound for malrotation in expert hands was ninety-four percent sensitive and one hundred percent specific in a 2021 meta-analysis. For Hirschsprung disease, suction rectal biopsy was ninety-three percent sensitive and ninety-eight percent specific, anorectal manometry ninety-one percent sensitive, and the contrast enema only seventy percent sensitive. Magnetic resonance enterography detected active ileal Crohn disease with eighty-four percent sensitivity and ninety-seven percent specificity. At the bottom of the list are breath tests judged against culture of small-bowel fluid: the glucose test was about fifty-five percent sensitive and eighty-three percent specific, and the lactulose test about forty-two percent sensitive and seventy-one percent specific. And in one careful study, in twenty-two of twenty-five positive lactulose tests, about eighty-eight percent, the test meal had already reached the cecum when hydrogen rose, so the test was measuring colonic fermentation, not overgrowth. Calprotectin, for comparison, has a sensitivity above ninety percent for pediatric inflammatory bowel disease, with a specificity of seventy to ninety percent.

Four numbers from the block change what happens next to a real child. First, biopsying only visible lesions misses about half of histological disease in children. Second, if calprotectin is used to decide who is scoped, sixty-five of every hundred children are scoped instead of all hundred, at the cost of about eight percent of diagnoses being delayed. Third, adult criteria labeled about two thirds of children abnormal on esophageal manometry, and adult anorectal criteria called about forty-six percent of healthy children dyssynergic. Fourth, a single-gene cause explains about three percent of all pediatric inflammatory bowel disease, but between thirteen and forty-one percent of cases that begin in infancy. The translation is to biopsy normal mucosa, let calprotectin decide who is scoped, never apply an adult motility threshold to a child without saying so, and sequence by age of onset rather than by severity. And always know the denominator behind a number: who was tested, against what, and at what age.

Part six. Five difficult cases.

Return to the three-year-old boy with four loose stools a day for eight weeks, undigested food, no blood, no stools at night, and weight and height on the fiftieth centile, unchanged since infancy. His hemoglobin, C-reactive protein and albumin are normal. His calprotectin is three hundred and eighty, and he had ibuprofen twice last week for a cold. A stool panel found norovirus. Healthy children under four run calprotectin values of one hundred to two hundred and fifty micrograms per gram, and anti-inflammatory drugs and infection push the value higher. Norovirus is found in about four percent of healthy American children under five, so its presence is a detection, not an explanation. Normal growth and no alarm features point to functional diarrhea of childhood under the Rome criteria. But normal blood tests do not exclude inflammatory bowel disease; half of children with mild ulcerative colitis had all four routine tests normal at diagnosis. The answer is therefore not reassurance alone and not a scope today. Stop the anti-inflammatory drugs, repeat the calprotectin on a formed sample in four to six weeks, and scope only if it stays high or growth falters.

The nine-year-old girl has post-meal bloating and pain around the umbilicus, with normal growth. Her fructose breath test, using twenty-five grams, showed a hydrogen rise of forty parts per million with bloating, and her lactulose test showed a rise of twenty-two at seventy-five minutes and was labeled overgrowth. Coeliac serology, calprotectin and blood count are normal. A twenty-five-gram fructose load is positive in about fifty-nine percent of healthy children, so the first result describes physiology. The lactulose result most likely reflects arrival of the sugar in the colon, and the lactulose test is only about forty-two percent sensitive and seventy-one percent specific against culture anyway. The Rome criteria allow a positive diagnosis after appropriate evaluation rather than exhaustive exclusion. The diagnosis is functional abdominal pain with bloating. She should not receive antibiotics for overgrowth, and she needs explanation and sensible dietary advice rather than long elimination diets.

The seven-month-old boy with Down syndrome passed meconium at fifty-two hours, has been constipated since the second week of life and has had two episodes of abdominal distension. His contrast enema showed no clear transition zone, and anorectal manometry while awake did not show the RAIR. A contrast enema is only about seventy percent sensitive for Hirschsprung disease, so a normal enema does not exclude it. The RAIR is present from birth, even in preterm infants, so its absence is a real signal. Hirschsprung disease is also more common in Down syndrome, occurring in just under one percent in the large American cohorts. The next step is a suction rectal biopsy, taking two or three samples at least two centimeters above the dentate line, with calretinin staining, which highlights the nerve fibers that are absent in aganglionosis. Do not repeat the manometry first.

The fourteen-year-old girl has a height z-score of minus two point one, down from minus zero point eight at age ten, and delayed puberty. Her tissue transglutaminase IgA is twelve times the

upper limit of normal, her endomysial antibody is positive on a second sample, and her total IgA is normal. Her family declines endoscopy, and next year they move to an adult service in North America. Under the 2020 ESPGHAN criteria she can be diagnosed without biopsy; the positive predictive value of that pathway was ninety-nine point seven five percent. About one in fourteen short children turns out to have coeliac disease, and her falling height z-score is itself malnutrition by trajectory. If she has bone densitometry, it should be reported as a z-score adjusted for her height, never as an adult T-score. The decision is to diagnose without biopsy, start a gluten-free diet, document both blood results clearly, and explain the decision in the transition letter, because her new service may expect a biopsy.

Finally, the fourteen-month-old boy has had bloody diarrhea since nine months of age, a perianal fistula and recurrent skin abscesses. His weight-for-length z-score is minus two point four and his mid-upper arm circumference z-score is minus two point two. His calprotectin is above eighteen hundred and his albumin is twenty-six grams per liter. This is very-early-onset inflammatory bowel disease, and a single-gene cause is found in thirteen to forty-one percent of children whose disease starts in infancy; perianal and skin disease raise the odds further. Before sequencing, screen the immune system: a blood count, immunoglobulins, lymphocyte subsets, and the dihydrorhodamine test, called DHR, which detects chronic granulomatous disease, a defect in how neutrophils kill bacteria. Then send trio genome or exome sequencing, testing the child and both parents together. The reason this matters is that only defects of blood-forming cells are cured by hematopoietic stem cell transplantation. By the 2014 malnutrition criteria he has moderate malnutrition. The plan is the immunology screen and trio sequencing on day one, upper endoscopy and ileocolonoscopy under general anesthesia with every segment in its own jar, and a nutrition plan now.

Part seven. Twenty pearls.

From the first half of the block, carry these ten. Alarm features were about as common in functional pain as in organic disease, sixty-one percent against fifty-six percent, so they target tests rather than diagnoses. Height velocity was already reduced in eighty-eight percent of children before Crohn disease was diagnosed. Half of children with mild ulcerative colitis had all four routine blood tests normal at diagnosis. Healthy infants run calprotectin values of one hundred to two hundred and fifty, settling after age three or four. Elastase is falsely low in watery stool, so repeat it on a formed sample. Forty-one percent of healthy infants carry Clostridioides difficile, fourteen percent with toxin-producing strains, so do not test under twelve months. A positive breath test means a hydrogen rise of at least twenty parts per million or methane of at least ten; stop antibiotics four weeks before, but not proton pump inhibitors. Fructose tests are positive in eighty-eight percent of infants and fifty-nine percent of healthy children. Biopsying only visible lesions misses about half of histological disease. And basal

plasmacytosis, meaning plasma cells crowded at the base of the mucosa, is found in sixty-three percent of first attacks of ulcerative colitis and six percent of infectious colitis.

From the second half, carry these ten. The normal pediatric bowel wall is no thicker than about one point nine millimeters, and the adult cut-off of three millimeters misses inflamed ilea in small children. Magnetic resonance enterography is eighty-four percent sensitive and ninety-seven percent specific for active ileal Crohn disease, and gadolinium should be added for perianal disease. Capsule retention was two point six percent overall in children and five point two percent in known inflammatory bowel disease; a patency capsule lowers that risk but does not abolish it. Non-acid reflux makes up forty-five to eighty-nine percent of pediatric reflux episodes, so pH monitoring alone misses it. Adult Chicago criteria labeled about two thirds of children abnormal, and the rate of esophageal spasm fell from fourteen percent to one percent with age adjustment. The RAIR is a drop in sphincter pressure of at least fifteen percent, and when it is absent the next step is a suction biopsy with calretinin. On gastric emptying scintigraphy, more than ten percent retained at four hours is delayed, and two-hour studies miss about half of delayed emptying. Single-gene inflammatory bowel disease accounts for about three percent overall, seven point eight percent under six years, and thirteen to forty-one percent with onset in infancy. Under the 2014 malnutrition criteria one indicator suffices, but height-for-age counts only at minus three, whereas the World Health Organization calls minus two stunting. And mid-upper arm circumference predicted mortality better than body mass index, with an area under the curve of zero point seven three against zero point five eight, and it is not inflated by edema.

Part eight. What we still do not know.

The block left a clear list of gaps. There are no agreed calprotectin cut-offs for children under four, which drives over-referral of healthy toddlers; age-banded reference ranges with a single assay would help. There are no validated pediatric breath test doses, and a fructose test that most healthy toddlers fail is not a useful test. There are no normal pH and impedance values in children, so every abnormal study is a judgment call. There are no age-specific thresholds for high-resolution esophageal or anorectal manometry. No society sets a weight floor for balloon enteroscopy. There are no age-specific cut-offs for bowel wall thickness on ultrasound, and the adult three millimeters misses very-early-onset disease. There is no pediatric version of the global adult malnutrition criteria. And the PEnQuIN indicators have not yet been tested against clinical outcomes, which is why forty-five of the forty-nine standards remain conditional.

What to do differently in clinic.

Here are the habits this block should leave behind. Plot every child as z-scores over time, and act on a fall of one z-score rather than on a single centile. Read calprotectin with age, drugs

and infection in mind, and repeat it before scoping a thriving toddler. Treat a stool test hit or a positive breath test as a finding to explain, not as a diagnosis. Biopsy normal mucosa by protocol, put each segment in its own jar, and ask the pathologist for peak eosinophil counts and the field area used. Choose ultrasound and magnetic resonance over computed tomography, and age-appropriate cut-offs over adult ones. Say the words adult threshold out loud whenever you apply one to a child's motility study. Sequence by age of onset and red flags, after an immunology screen, and plan to re-analyze negative results. And audit your own endoscopy against PEnQuIN: preparation in at least eighty percent, the cecum in at least ninety, the ileum in at least eighty-five, and every adverse event recorded. A test is only as good as the question it was ordered to answer.